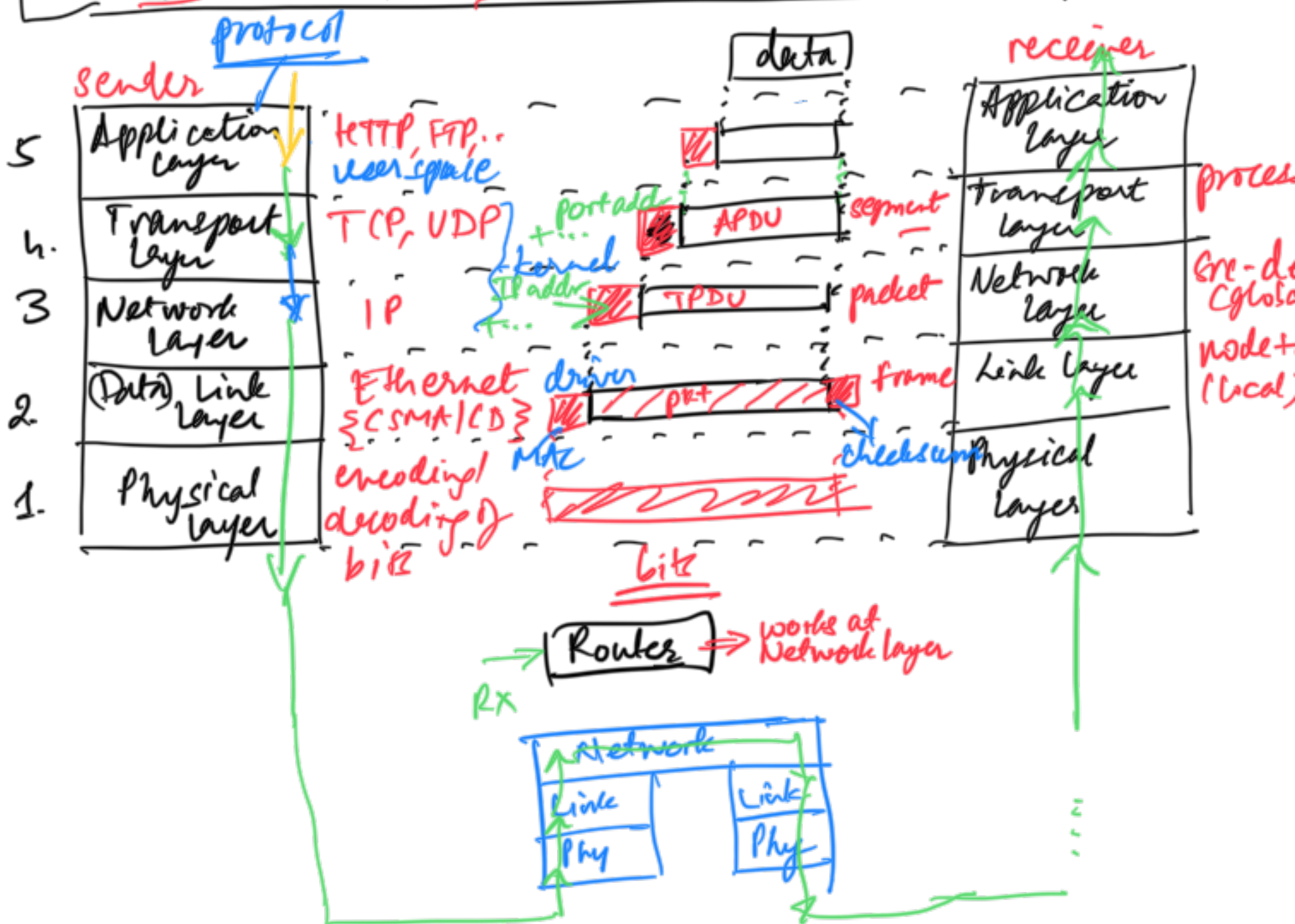


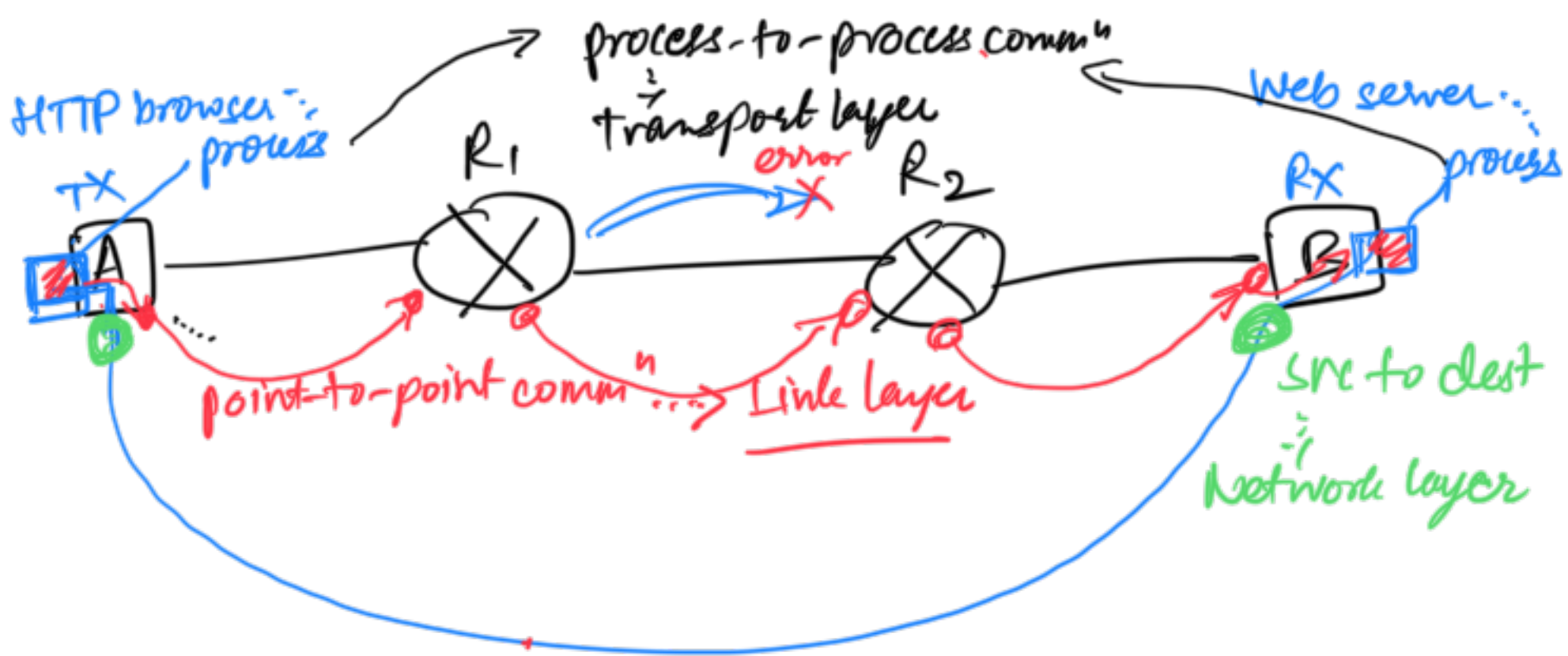
Lec 0 - Communication models

- Divide the commⁿ system functions into a set of layers
 - ① Each layer performs a defined set of functions
 - ② Abstraction
 - ③ Each layer is implemented using a set of protocols

→ Internet standards

* TCP/IP model | Internet model | Networking Stack





* Transport layer

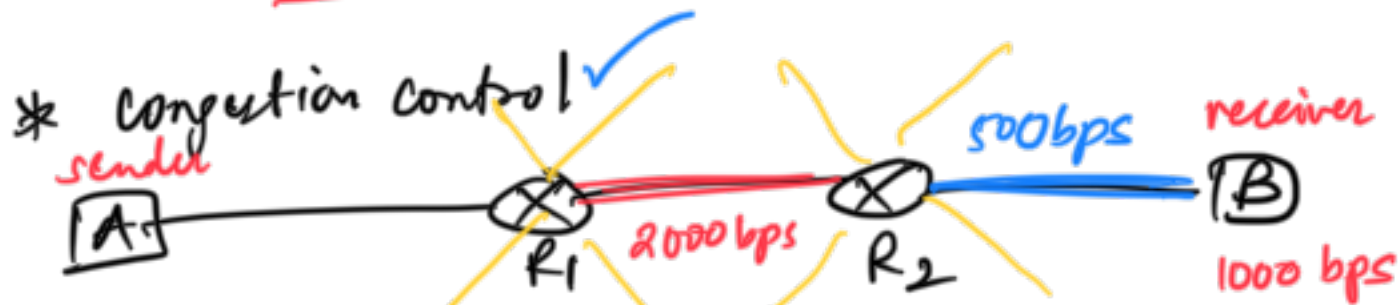
* process-to-process delivery of segments

* port-addressing

* flow control



* Error control RX tests TX, speed at which TX should send
 error detection & correction or retransmission
 (extra bits) checksums



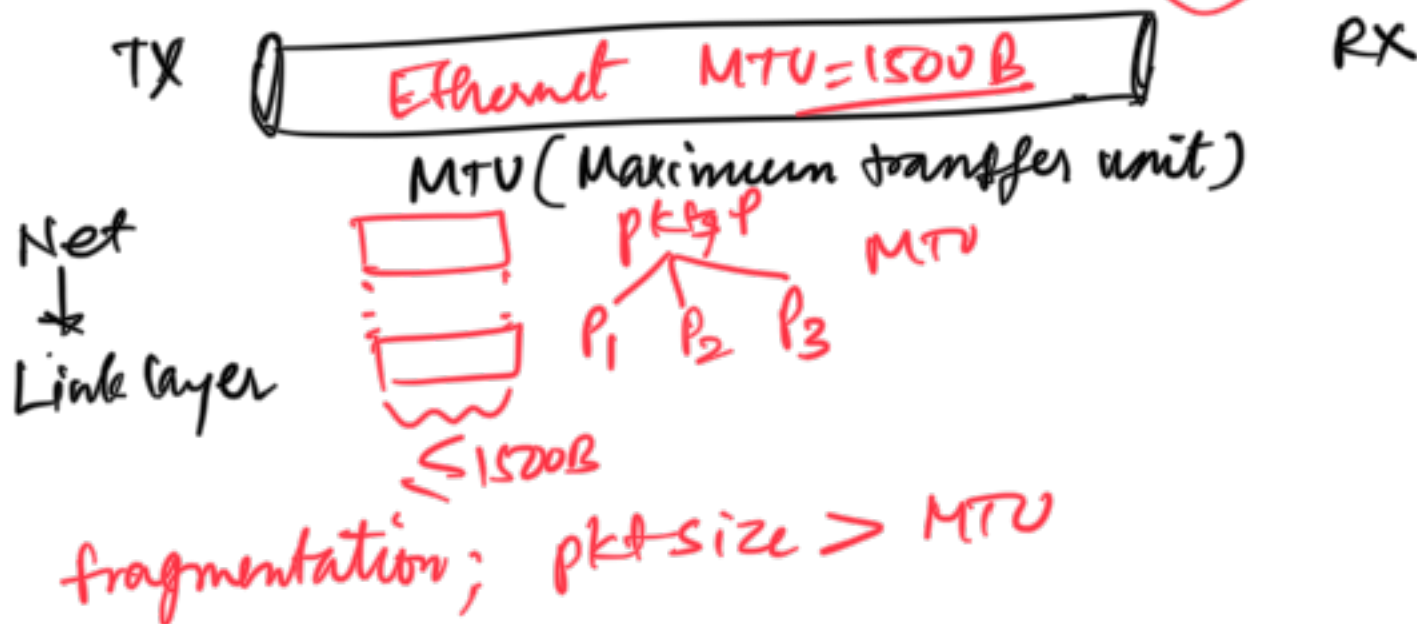
* Segmentation & reassembly

* Network layer

- src to dest delivery of pkts
IP addresses

- routing

- fragmentation & reassembly



* Link layer

- node to node delivery
MAC addresses

- flow control

- error control

- access control

↳ decide who sends data, when? & for how much time??

Internetworking devices [will discuss details later in the layer at which they work course]
Physical layer

HUBS, REPEATERS

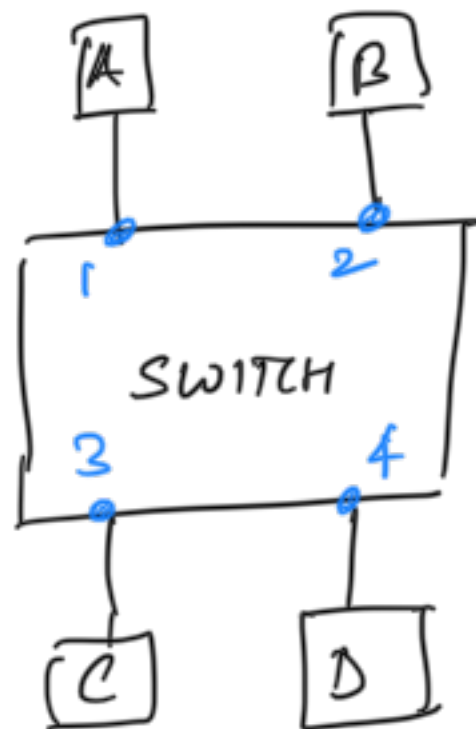
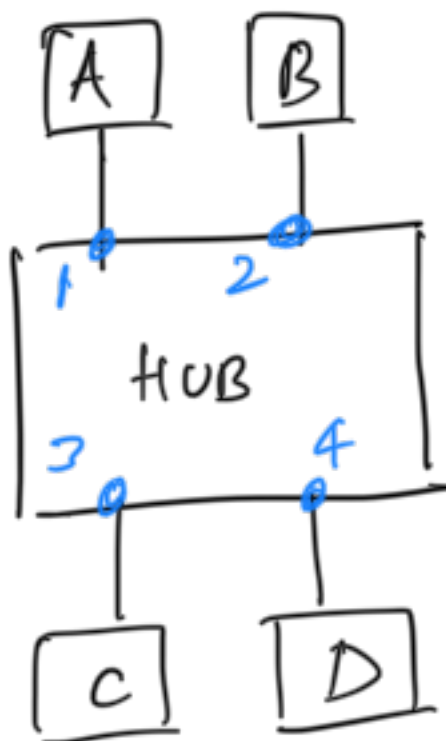
SWITCH, BRIDGE Data link layer

ROUTER

Network layer

GATEWAYS

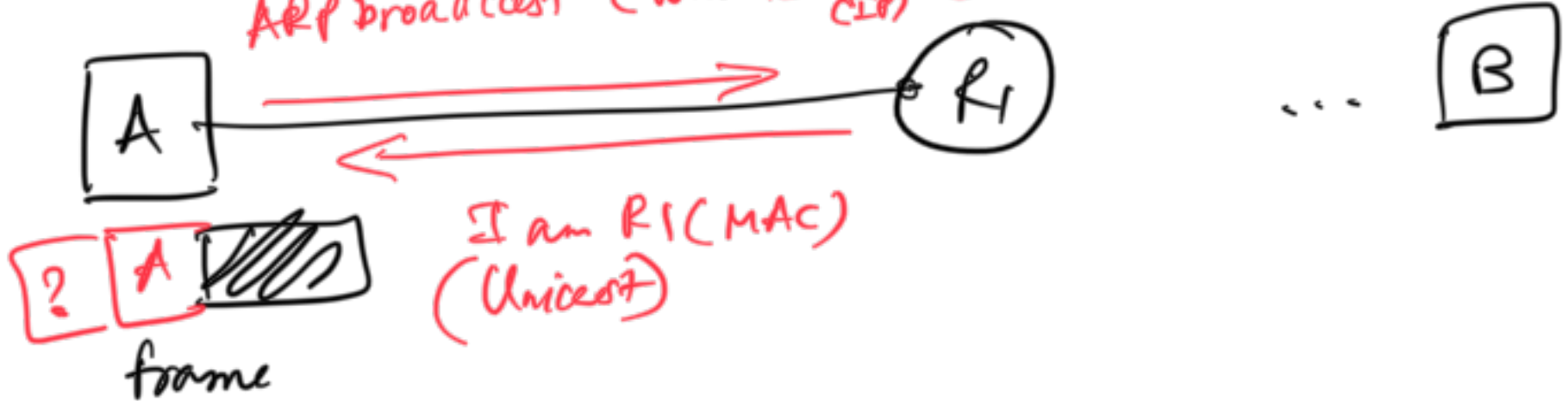
Transport layer & above



ARP Address Resolution Protocol

* maps IP addr to MAC addr

ARP broadcast (Who is R_1 ?)
(IP)



RARP (Reverse ARP)

* maps MAC addr to IP addr

- given that a host can be assigned IP address dynamically, RARP is used to ask for an IP address when a host boots up or the lease expires